2 A car of mass 1800 kg accelerates from rest on a horizontal surface.

The graph in Fig. 2.1 shows how the driving force of the car's engine F_E, and the resistive force on the car F_R, vary with the distance d travelled by the car.

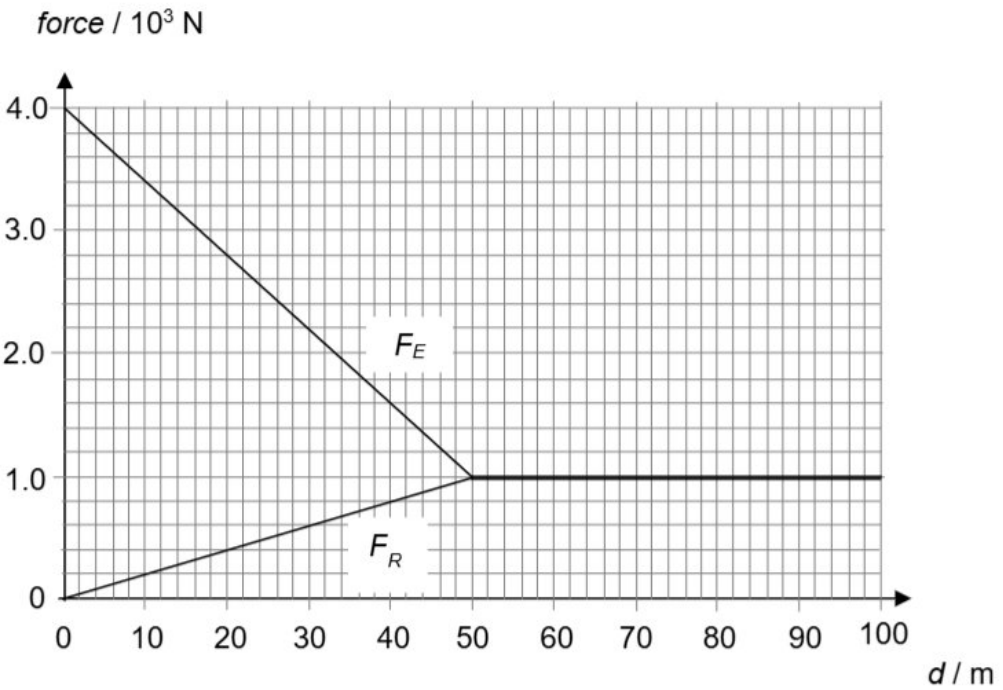


Fig. 2.1

(a) State what is represented by the area under the line F_E.

.....

..... [1]

(b) **(i)** Calculate the maximum speed of the car.

maximum speed = ... m s^{-1} [3]

(ii) Calculate the power output of the engine when the car is travelling at its maximum speed.

power output = ... W [2]

(c) The mass of the car is increased when more passengers enter the car.

The power output of the motor and the resistive forces on the car remain unchanged.

Explain why the car now requires a longer time to reach the same maximum speed as that in part **(b)(i)**.

.....

..... [1]